

Main Project 01 – Weather Data API

Introduction

In this Open Weather Map, which is used to get the weather on respective locations, It contains only a weather map page. We are going to perform the load test, stress test, spike test, endurance test, and volume test here.

- Web application at [https://www. openweathermap.org.com](https://www.openweathermap.org.com)

Test Setup

Weather Data API

1. [https://www.openweathermap.org.com /](https://www.openweathermap.org.com/) - Weather Page

Test-Setup

1. From blaze meter to get the data from website.
2. Using Synchronizing Timer to integrate no of user requests.
3. Using Random Controller to process the requests on randomly.
4. Using Loop Controller in infinite mode to repeat the activity.
5. The above features are integrating with multiple thread group by duration to stop specify thread lifetime according to the thread group.

Environment Setup

1. **Processor** - 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz
2. **RAM** – 8GB

Jenkins

It integrated with Jenkin's.

GitHub

1. [Performance Testing/Weather API load test.jmx at master · senthilraja29/Performance Testing \(github.com\)](#)
2. [Performance Testing/Weather-Data - Endurance Test.jmx at master · senthilraja29/Performance Testing \(github.com\)](#)
3. [Performance Testing/Weather-Data - Scalability Test.jmx at master · senthilraja29/Performance Testing \(github.com\)](#)
4. [Performance Testing/Weather-Data - Spike Test.jmx at master · senthilraja29/Performance Testing \(github.com\)](#)
5. [Performance Testing/Weather-Data - Stress Test.jmx at master · senthilraja29/Performance Testing \(github.com\)](#)

Environment

Running Environment JMeter.

Test Results

Load Test

In this load test to perform 1500 concurrent users hitting at 30 mints. After conducting load test around 68 thousand users hit the website within 30 mints, we received **4.35%** of error.

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Weather Loc...	6879	37041	502	70250	12394.70	4.35%	3.3/sec	108.11	0.39	33074.9
Weather Loc...	6781	37173	533	58152	12214.03	5.06%	3.3/sec	105.77	0.39	32838.4
Weather Loc...	6857	36896	504	62730	12615.66	4.45%	3.3/sec	107.63	0.39	33043.0
Weather Loc...	6880	37142	553	68805	12308.91	4.27%	3.3/sec	108.20	0.39	33098.6
Weather Loc...	6832	36969	523	63316	12419.84	4.92%	3.3/sec	106.70	0.39	32882.3
Weather Loc...	6820	36911	536	68968	12495.09	4.75%	3.3/sec	106.72	0.39	32940.3
Weather Loc...	6938	37256	567	58852	12191.81	4.27%	3.4/sec	109.08	0.40	33103.0
Weather Loc...	6800	37252	551	57484	12290.64	4.43%	3.3/sec	106.79	0.39	33050.9
Weather Loc...	6754	37077	524	67748	12316.34	4.43%	3.3/sec	106.00	0.39	33045.0
Weather Loc...	6837	37039	559	59675	12450.57	4.34%	3.3/sec	107.41	0.39	33077.0
TOTAL	68378	37076	502	70250	12370.98	4.52%	33.3/sec	1072.23	3.92	33015.7

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

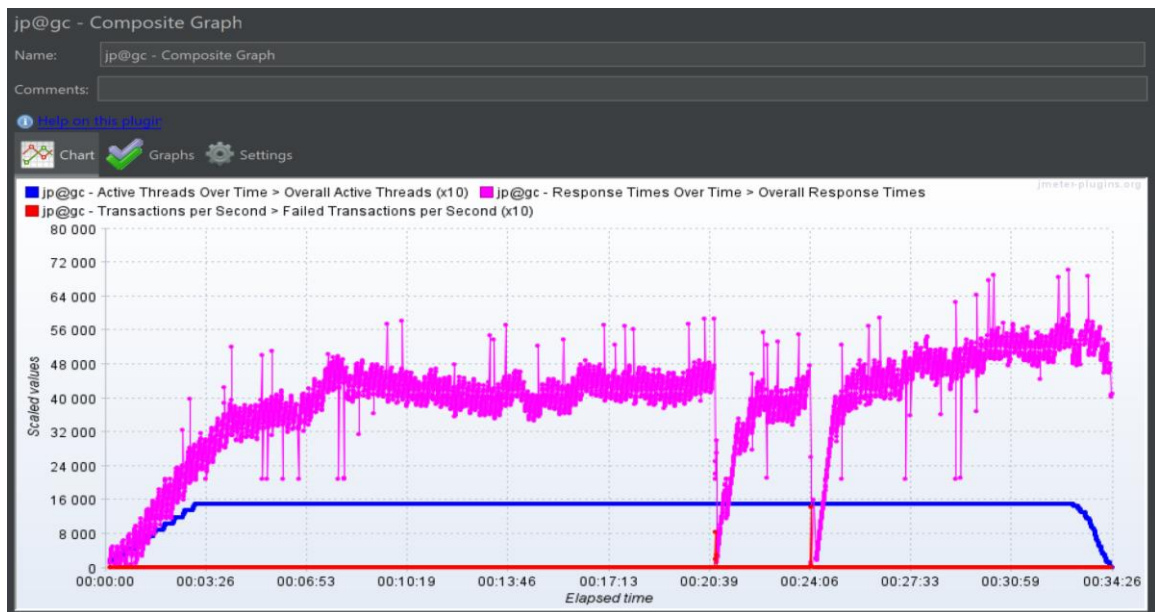
Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Weather L...	6879	37041	40575	49448	51965	54977	502	70250	4.35%	3.3/sec	108.11	0.39
Weather L...	6781	37173	40598	49634	52420	55202	533	58152	5.06%	3.3/sec	105.77	0.39
Weather L...	6857	36896	40537	49589	52260	55078	504	62730	4.45%	3.3/sec	107.63	0.39
Weather L...	6880	37142	40486	49738	52538	55079	553	68805	4.27%	3.3/sec	108.20	0.39
Weather L...	6832	36969	40447	49597	52442	55190	523	63316	4.92%	3.3/sec	106.70	0.39
Weather L...	6820	36911	40492	49586	52295	55272	536	68968	4.75%	3.3/sec	106.72	0.39
Weather L...	6938	37256	40563	49694	52502	55244	567	58852	4.27%	3.4/sec	109.08	0.40
Weather L...	6800	37252	40595	49645	52203	54866	551	57484	4.43%	3.3/sec	106.79	0.39
Weather L...	6754	37077	40621	49566	52149	55152	524	67748	4.43%	3.3/sec	106.00	0.39
Weather L...	6837	37039	40455	49826	52376	55159	559	59675	4.34%	3.3/sec	107.41	0.39
TOTAL	68378	37076	40539	49625	52325	55144	502	70250	4.52%	33.3/sec	1072.23	3.92



Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Weather Location 01	12605.86987	36936.32	59666	0
Weather Location 02	12792.47112	36782.2	62730	0
Weather Location 03	12408.01626	37005.84	67747	0
Weather Location 04	12605.86647	36850.51	63148	0
Weather Location 05	12694.64605	36787.2	67543	0
Weather Location 06	12419.05855	37063.39	68797	0
Weather Location 07	12483.26587	37133.55	57464	0
Weather Location 08	12523.36058	36952.24	70249	0
Weather Location 09	12481.08154	37019.06	58142	0
Weather Location 10	12338.95579	37157.37	58843	0
Grand Total	12535.65793	36968.92	70249	0

Observations:

1. **4.35%** errors encountered during load test around 68378 transactions.
2. Average Response time **37076 ms**
3. 95th percentile fall under – **51965 ms**
4. Throughput (requests per second) - **33.3/ sec**
5. Network performance (Max latency) – **70249 ms**
6. Network performance (Average latency) – **36968.92 ms**

Stress Test

In this stress test, which is conducted by stepping thread groups to perform 5000 users holds for 1200 seconds. In this load, 500 threads are gradually added by each 30-second interval.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

ting\performance testing\Projects\Main 01\Stress test\Weather-Data - Stress Test.cs

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Kuala Lump...	23060	23227	0	149295	29035.49	96.33%	14.4/sec	38.14	1.84	2709.5
London Wea...	23034	23376	0	149306	29020.36	96.35%	14.4/sec	38.03	1.85	2705.0
Tokyo Weat...	23093	23615	0	149192	28898.77	96.18%	14.4/sec	38.36	1.90	2721.3
Mumbai We...	23045	23337	0	149419	29084.22	96.64%	14.4/sec	37.38	1.84	2657.2
Rome Weat...	23291	23027	0	149398	28570.21	96.34%	14.6/sec	38.56	1.85	2712.2
Hanoi Weat...	22895	23060	0	149449	28916.76	96.30%	14.3/sec	37.87	1.84	2709.9
Delhi Weather	22915	23615	0	149294	29114.63	96.39%	14.3/sec	37.69	1.86	2694.2
Paris Weather	23069	23151	0	149403	28819.08	96.43%	14.4/sec	37.98	1.83	2697.1
Berlin Weath...	23011	22987	0	149276	28854.38	96.45%	14.4/sec	37.87	1.81	2696.3
Seattle Weat...	22737	23192	0	149399	28836.94	96.35%	14.2/sec	37.54	1.82	2704.7
TOTAL	230150	23258	0	149449	28915.90	96.38%	143.9/sec	379.41	18.45	2700.7

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

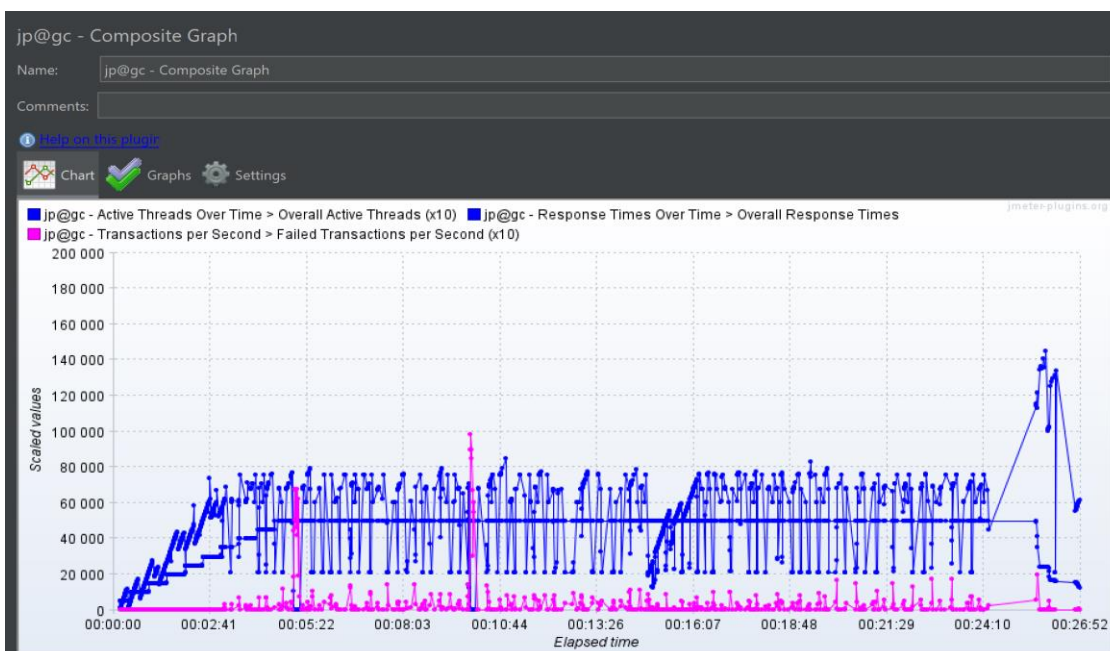
Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/se
Kuala Lu...	23060	23227	21026	68252	75603	101847	0	149295	96.33%	14.4/sec	38.14	1.84
London W...	23034	23376	21029	68241	75598	101877	0	149306	96.35%	14.4/sec	38.03	1.85
Tokyo We...	23093	23615	21031	68143	75596	101679	0	149192	96.18%	14.4/sec	38.36	1.90
Mumbai ...	23045	23337	21027	68247	75595	101898	0	149419	96.64%	14.4/sec	37.38	1.84
Rome We...	23291	23027	21029	68124	75593	101469	0	149398	96.34%	14.6/sec	38.56	1.85
Hanoi We...	22895	23060	21023	68090	75591	101689	0	149449	96.30%	14.3/sec	37.87	1.84
Delhi Wea...	22915	23615	21031	68242	75601	101935	0	149294	96.39%	14.3/sec	37.69	1.86
Paris Wea...	23069	23151	21028	68096	75586	101876	0	149403	96.43%	14.4/sec	37.98	1.83
Berlin W...	23011	22987	21024	68120	75595	101878	0	149276	96.45%	14.4/sec	37.87	1.81
Seattle W...	22737	23192	21026	68190	75598	101748	0	149399	96.35%	14.2/sec	37.54	1.82
TOTAL	230150	23258	21028	68170	75595	101788	0	149449	96.38%	143.9/sec	379.41	18.45



Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Berlin Weather	29508.31256	16235.36	149276	0
Delhi Weather	29884.69703	16787.71	149294	0
Hanoi Weather	29663.90003	16509.03	149449	0
Kuala Lumpur Weather	29768.48033	16515.59	149295	0
London Weather	29808.51005	16585.9	149306	0
Mumbai Weather	29871.1098	16620.78	149419	0
Paris Weather	29557.56262	16336.26	149403	0
Rome Weather	29470.63428	16315.56	149398	0
Seattle Weather	29591.25118	16531.73	149399	0
Tokyo Weather	29937.57524	16995.87	149192	0
Grand Total	29707.2838	16543.11	149449	0

Observations:

1. During stress test error occurred which is **96.33%** (504).
2. Average Response time **23227ms**
3. 95th percentile falls under - **75595ms**
4. Throughput (requests per second) – **143.9ms**
5. Network performance (Max latency) – **149449ms**
6. Network performance (Average latency) – **16543.11ms**

Spike Test

In this spike test, which is conducted by open model thread group to perform 1000 users interact within 40 seconds. In this test started by 10 seconds and completed by 60 seconds and maintained 1 minute of transaction rate up to 100. Load rate will be 400/seconds. Around 26367 samples were transitioned.

Summary Report

Name:

Summary Report

Comments:

- Write results to file / Read from file

Filename

sting\performance testing\Projects\Main 01\Spike test\Weather-Data - Spike Test.cs

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Hanoi Weat...	2709	36613	0	99703	22926.30	76.97%	15.2/sec	154.58	1.71	10389.1
Seattle Weat...	2614	36666	0	99782	23192.13	76.09%	14.7/sec	153.09	1.71	10673.3
Kuala Lump...	2654	36983	0	99634	22684.38	78.15%	14.6/sec	142.82	1.55	10004.1
Paris Weather	2640	36391	0	100056	22603.69	77.61%	14.5/sec	143.97	1.58	10187.6
Rome Weat...	2658	36644	0	99824	22721.85	77.43%	14.8/sec	148.49	1.63	10246.2
Tokyo Weat...	2583	36340	0	100080	23290.34	76.35%	14.4/sec	149.32	1.66	10585.4
Berlin Weath...	2582	36621	0	100021	23008.55	76.96%	14.4/sec	146.02	1.61	10390.5
Mumbai We...	2623	36965	0	99691	22619.96	76.52%	14.6/sec	150.11	1.67	10532.2
Delhi Weather	2645	36634	0	99744	23423.86	75.77%	14.6/sec	153.01	1.71	10768.1
London Wea...	2659	37088	0	99721	23114.58	77.85%	15.0/sec	147.99	1.61	10111.9
TOTAL	26367	36696	0	100080	22960.42	76.97%	144.5/sec	1466.13	16.19	10387.6

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Hanoi We...	2709	36613	34985	70798	74787	93913	0	99703	76.97%	15.2/sec	154.58	1.71
Seattle W...	2614	36666	34935	70895	76977	95267	0	99782	76.09%	14.7/sec	153.09	1.71
Kuala Lu...	2654	36983	35084	70323	73400	90768	0	99634	78.15%	14.6/sec	142.82	1.55
Paris Wea...	2640	36391	34371	70020	75733	93272	0	100056	77.61%	14.5/sec	143.97	1.58
Rome We...	2658	36644	34820	70597	76016	92368	0	99824	77.43%	14.8/sec	148.49	1.63
Tokyo We...	2583	36340	34447	70738	75782	94360	0	100080	76.35%	14.4/sec	149.32	1.66
Berlin We...	2582	36621	34636	70797	76404	94640	0	100021	76.96%	14.4/sec	146.02	1.61
Mumbai ...	2623	36965	34964	70792	75684	91007	0	99691	76.52%	14.6/sec	150.11	1.67
Delhi Wea...	2645	36634	34874	71078	77444	94586	0	99744	75.77%	14.6/sec	153.01	1.71
London W...	2659	37088	36162	70894	76783	94316	0	99721	77.85%	15.0/sec	147.99	1.61
TOTAL	26367	36696	34935	70755	75962	93864	0	100080	76.97%	144.5/sec	1466.13	16.19

jp@gc - Composite Graph

Name: jp@gc - Composite Graph

Comments:

[Help on this plugin](#)



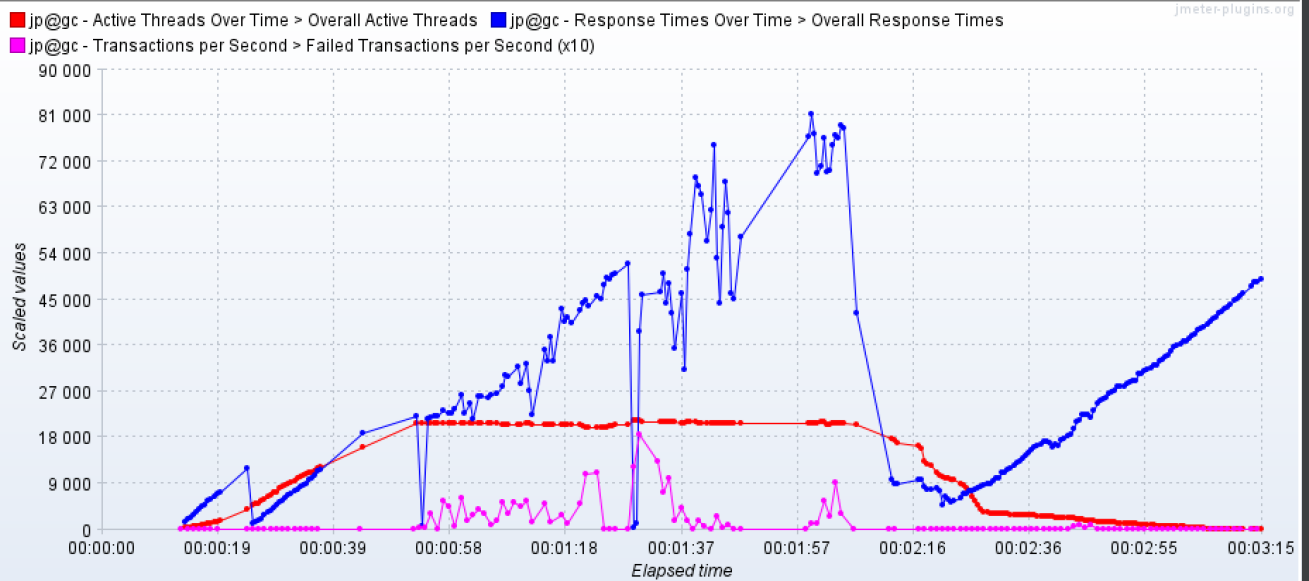
Chart



Graphs



Settings



Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Weather Location 01	16062.57822	6151.167	90978	0
Weather Location 02	16318.79822	6212.018	98293	0
Weather Location 03	16614.22389	6433.016	98488	0
Weather Location 04	16191.03016	6220.371	94810	0
Weather Location 05	16600.48687	6402.019	96027	0
Weather Location 06	17228.99993	6700.496	94657	0
Weather Location 07	16372.71811	6333.266	94448	0
Weather Location 08	16583.74745	6413.286	94680	0
Weather Location 09	16592.10309	6331.801	94598	0
Weather Location 10	16715.90461	6464.063	95380	0
Grand Total	16528.25506	6365.927	98488	0

Observations:

1. During Spike test error occurred which is **76.97%**
2. Average Response time **36613ms**
3. 95th percentile Response falls under – **75962ms**
4. Throughput (requests per second) – **144.5/sec**
5. Network performance (Max latency) – **98488ms**
6. Network performance (Average latency) – **6365.92ms**

Endurance Test

In this endurance test, which is conducted by Concurrency thread group to perform 1500 users interact within 120 minutes. In this test, the full load started after 150 seconds and was completed by 1 hour 59 minutes. Each request is integrated with a uniform random timer to simulate real-world scenarios. All the requests will under synchronize timer for 1500 concurrent users.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

ormance testing\Projects\Main 01\Endurance test\Weather-Data - Endurance Test.c

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Hanoi Weat...	3874	52462	478	1414284	244431.88	14.95%	2.5/sec	36.43	1.18	15201.7
Paris Weather	3840	49245	481	1414194	233311.89	15.18%	2.4/sec	36.01	1.17	15157.9
Berlin Weath...	3753	47959	480	1414315	233938.71	14.04%	2.4/sec	35.62	1.14	15344.0
Delhi Weather	3795	55117	480	1414337	251491.68	15.60%	2.4/sec	35.44	1.15	15092.5
London Wea...	3734	49222	477	1414291	233426.77	15.13%	2.4/sec	35.04	1.14	15165.8
Kuala Lump...	3821	45885	485	1414428	227954.20	15.21%	2.4/sec	35.80	1.17	15143.6
Tokyo Weat...	3723	52342	478	1414281	242607.11	16.06%	2.4/sec	34.58	1.13	15014.2
Rome Weat...	3694	41902	477	1414102	215039.90	14.78%	2.3/sec	34.75	1.13	15207.3
Seattle Weat...	3729	52490	486	1414153	245617.72	14.51%	2.4/sec	35.24	1.13	15272.5
Mumbai We...	3722	56149	477	1414583	255006.75	16.63%	2.4/sec	34.36	1.13	14922.9
TOTAL	37685	50285	477	1414583	238604.89	15.21%	23.9/sec	353.25	11.46	15152.4

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

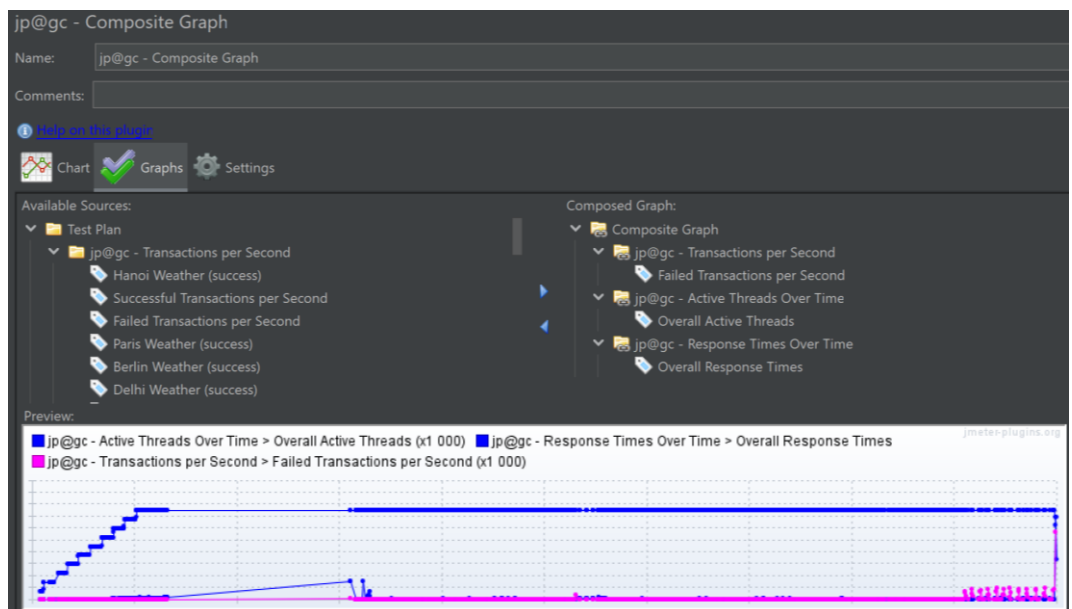
Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Hanoi We...	3874	52462	4448	22516	36552	1403508	478	1414284	14.95%	2.5/sec	36.43	1.18
Paris Wea...	3840	49245	4509	23551	35021	1393714	481	1414194	15.18%	2.4/sec	36.01	1.17
Berlin We...	3753	47959	4448	21043	35097	1403358	480	1414315	14.04%	2.4/sec	35.62	1.14
Delhi Wea...	3795	55117	4350	21095	36096	1393760	480	1414337	15.60%	2.4/sec	35.44	1.15
London W...	3734	49222	4288	22456	36778	1403447	477	1414291	15.13%	2.4/sec	35.04	1.14
Kuala Lu...	3821	45885	4314	21069	34087	1403052	485	1414428	15.21%	2.4/sec	35.80	1.17
Tokyo We...	3723	52342	4366	22873	36104	1403481	478	1414281	16.06%	2.4/sec	34.58	1.13
Rome We...	3694	41902	4290	21117	34170	1393408	477	1414102	14.78%	2.3/sec	34.75	1.13
Seattle W...	3729	52490	4439	23310	36377	1393944	486	1414153	14.51%	2.4/sec	35.24	1.13
Mumbai ...	3722	56149	4316	22692	35208	1403521	477	1414583	16.63%	2.4/sec	34.36	1.13
TOTAL	37685	50285	4378	22234	35445	1403105	477	1414583	15.21%	23.9/sec	353.25	11.46



Row Labels	Average of Latency	StdDev of Latency	Max of Latency	Min of Latency
Berlin Weather	7325.249589	19951.15	318988	0
Delhi Weather	8049.91476	26530.24	319094	0
Hanoi Weather	7787.664267	23360.33	319112	0
Kuala Lumpur Weather	7405.749395	21038.78	319028	0
London Weather	8522.767769	27819.53	319033	0
Mumbai Weather	7516.024743	22665.97	318982	0
Paris Weather	8399.214706	26413.96	319056	0
Rome Weather	7555.434517	21517.3	319080	0
Seattle Weather	7770.125381	22126	319083	0
Tokyo Weather	8202.242172	26422.45	318990	0
Grand Total	7854.124166	23933.1	319112	0

Observations:

1. During Endurance test error occurred which is **15.21%** (Response code 403 error)
2. Average Response time **50285ms**
3. 95th percentile Response times - **35445ms**
4. Throughput (requests per second) – **23.9/Sec**
5. Network performance (Max latency) – **319112ms**
6. Network performance (Average latency) – **7854.12ms**

Volume Test

In this volume test, which is conducted by the Ultimate thread group to perform 1500 users interact with 540 seconds. In this test, which was performed by various load actives 500, 900 and full load 1500. This full load 1500 users in 5.45 minutes. Each request is integrated with a uniform random timer and random controller to simulate real-world scenarios.

Requests	Executions				Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent	
Total	20464	97	0.47%	17216.10	1042	52033	14963.00	35085.90	36423.95	38510.98	36.43	625.03	18.06	
Berlin Weather	2028	13	0.64%	17628.94	1076	51035	15140.50	35183.40	36633.75	38507.30	3.61	61.93	1.79	
Delhi Weather	2035	15	0.74%	16869.13	1044	40990	14528.00	35040.20	36285.20	38345.72	3.64	62.33	1.80	
Hanoi Weather	2011	9	0.45%	17736.94	1100	41427	14956.00	35419.20	36640.00	38599.68	3.60	61.74	1.78	
Kuala Lumpur Weather	2100	6	0.29%	16962.92	1082	50027	14615.00	35109.70	36345.90	38393.29	3.75	64.50	1.86	
London Weather	2032	7	0.34%	17440.70	1109	41044	14672.00	35030.10	36271.40	38601.49	3.63	62.39	1.80	
Mumbai Weather	2044	8	0.39%	16894.38	1096	47658	14600.50	34819.50	36181.25	38413.65	3.65	62.70	1.81	
Paris Weather	2005	8	0.40%	16783.70	1086	41609	14390.00	35036.60	36451.50	38452.50	3.58	61.51	1.78	
Rome Weather	2050	15	0.73%	17216.87	1042	52033	14877.50	34820.80	36191.35	38683.88	3.65	62.48	1.80	
Seattle Weather	2099	9	0.43%	16994.88	1085	40883	14605.00	34730.00	36109.00	38483.00	3.75	64.42	1.86	
Tokyo Weather	2060	7	0.34%	17645.21	1082	51585	15100.50	35161.40	36843.50	38642.46	3.68	63.22	1.83	

Row Labels	StdDev of Latency	Average of Latency	Max of Latency
Berlin Weather	11804.83027	17485.80819	50121
Delhi Weather	11964.72956	16699.79558	40745
Hanoi Weather	12048.87671	17634.95624	41427
Kuala Lumpur Weather	11841.18639	16896.42429	50027
London Weather	11789.43048	17360.46998	40721
Mumbai Weather	11727.7024	16805.7456	47658
Paris Weather	11816.18291	16693.60998	41609
Rome Weather	12009.84374	17052.28244	51095
Seattle Weather	11699.97244	16898.04335	40883
Tokyo Weather	11982.33865	17566.37524	51584
Grand Total	11871.03256	17108.22435	51584

Observations:

1. During Volume test error occurred which is **0.43%**.
2. Average Response time **17216.16ms**
3. 95th percentile Response times – **36423.95ms**
4. Throughput (requests per second) – **36.43ms**

5. Network performance (Max latency) – **51584ms**
6. Network performance (Average latency) – **17108.22ms**

Conclusion

After conducting various tests from various thread groups, we will come to know

- **Load Test** - After performing load test we received **4.35%** of error around 68k transactions. Error starts after 20 mins requests processed.
- **Stress Test** - After performing stress test we received **96.33%** of error around 2.3lakh transactions. Error starts at starting stage itself and requests processed Latency Report shows 1.4 lakh is maximum duration takes.
(Non-HTTP response message: Network is unreachable: connect).
- **Endurance Test** - After performing Endurance test, we received **14.95%** of error. Major error occurs at non http respond.
- **Spike Test** - After performing Spike test, we received **76.75%** of error.
- **Volume Test** - After performing the volume test, an error occurred, which is **0.47%**.

Website can able to handle Volume and Load Tests. This Webpage unable to handle stress test, Endurance and spike test.

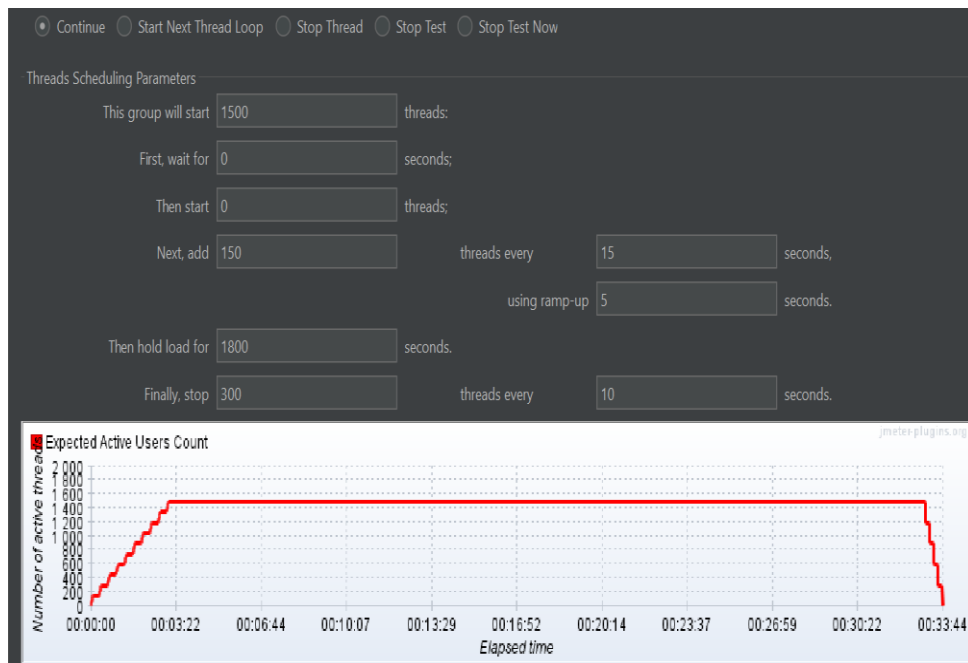
Appendix

Load Test



Configurations

- **Assertion** – No Assertion Integrated.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



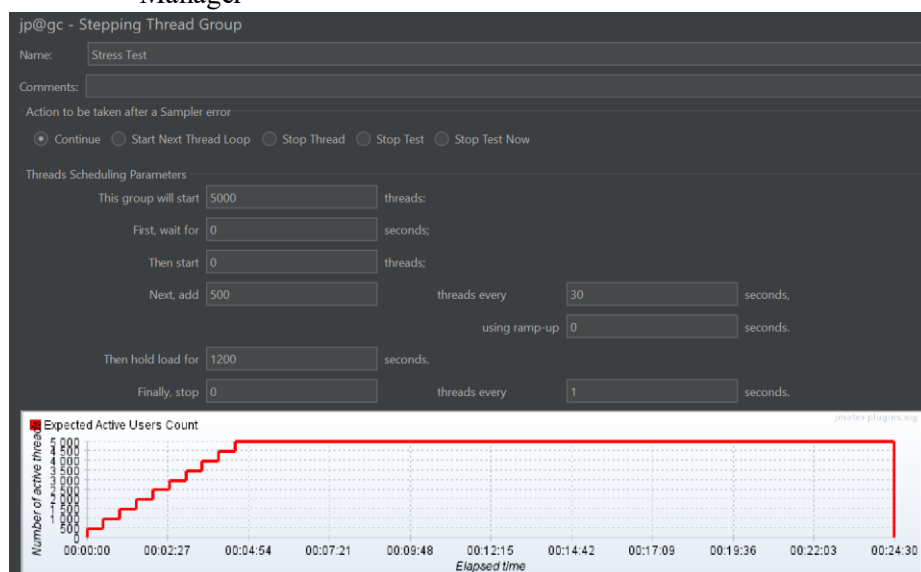
Stress Test



Weather-Data -
Stress Test.jmx

Configurations

- **Assertion** – No Assertion Added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



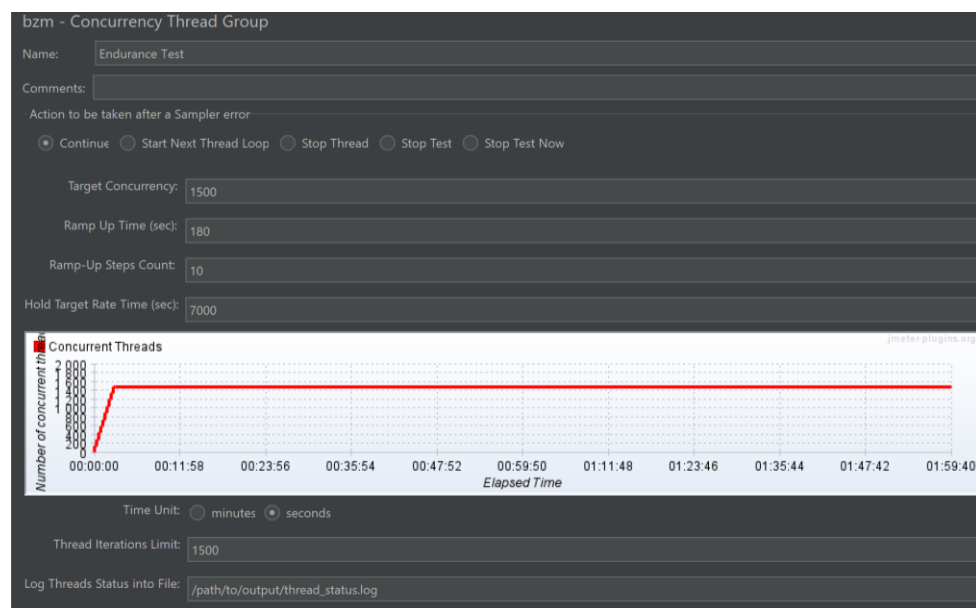
Endurance Test



Weather-Data -
Endurance Test.jmx

Configurations

- **Assertion** – No Assertion Integrated
- **Timer** - Synchronizing Timer, Uniform Random Timer
- **Thread Group** - Concurrency Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



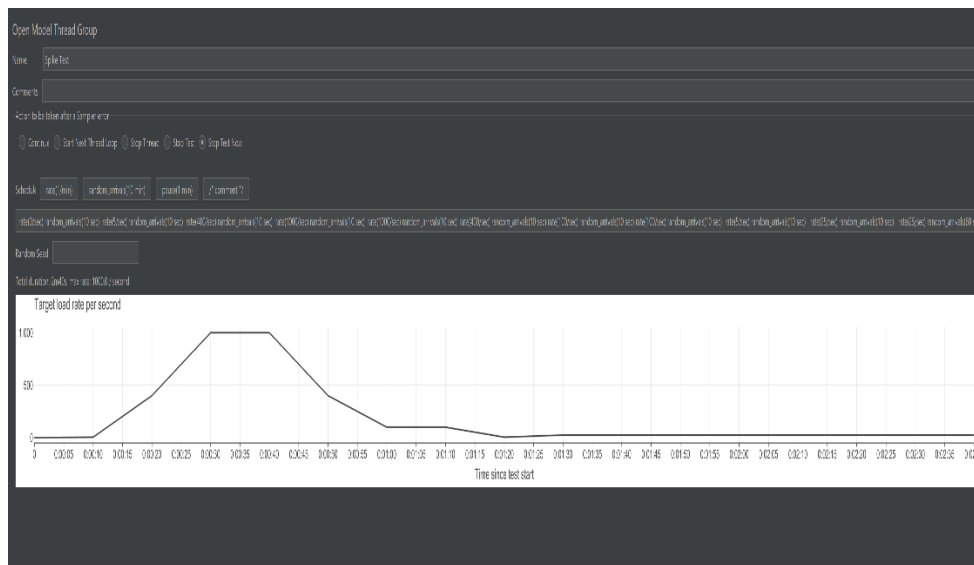
Spike Test



Weather-Data - Spike
Test.jmx

Configurations

- **Assertion** – No Assertion Integrated.
- **Timer** - Synchronizing Timer
- **Thread Group** – Open Model Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



Volume Test



Weather-Data -
Scalability Test.jmx

Configurations

- **Assertion** – No assertion added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Ultimate Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager

